**Department of Electronics and Computer Engineering**
**Institute of Engineering, Thapathali Campus, TU, Nepal**
**MSc in Informatics and Intelligent Engineering Program**

**Thesis Defense on**

# ADVERSARIAL ATTACKS ON MALWARE CLASSIFICATION MODELS

**Student:**

**Gita Gurung**
**Department of Electronics and Computer Engineering,**
**IOE Thapathali Campus,**
**078MSIISE008**

**Supervised by:**

**Er. Hari Prasad Pokhrel**
**Er. Umesh Kanta Ghimire**

**Date : APRIL, 2025**

# Outline

- Motivation
- Background
- Problem Statement
- Objectives of project
- Scope of project
- Potential Applications
- Literature Review
- Proposed Methodology
- Results
- Future Enhancement
- Conclusion
- References

# Motivation

- Adversarial instances pose a significant obstacle to the current state of malware detection and classification research.
- Proposing malware detectors or classifiers without taking adversarial attacks into account is a fruitless.
- Adversarial attacks challenge the limits of machine learning models, pushing researchers to develop better defenses and enhance security

# Background

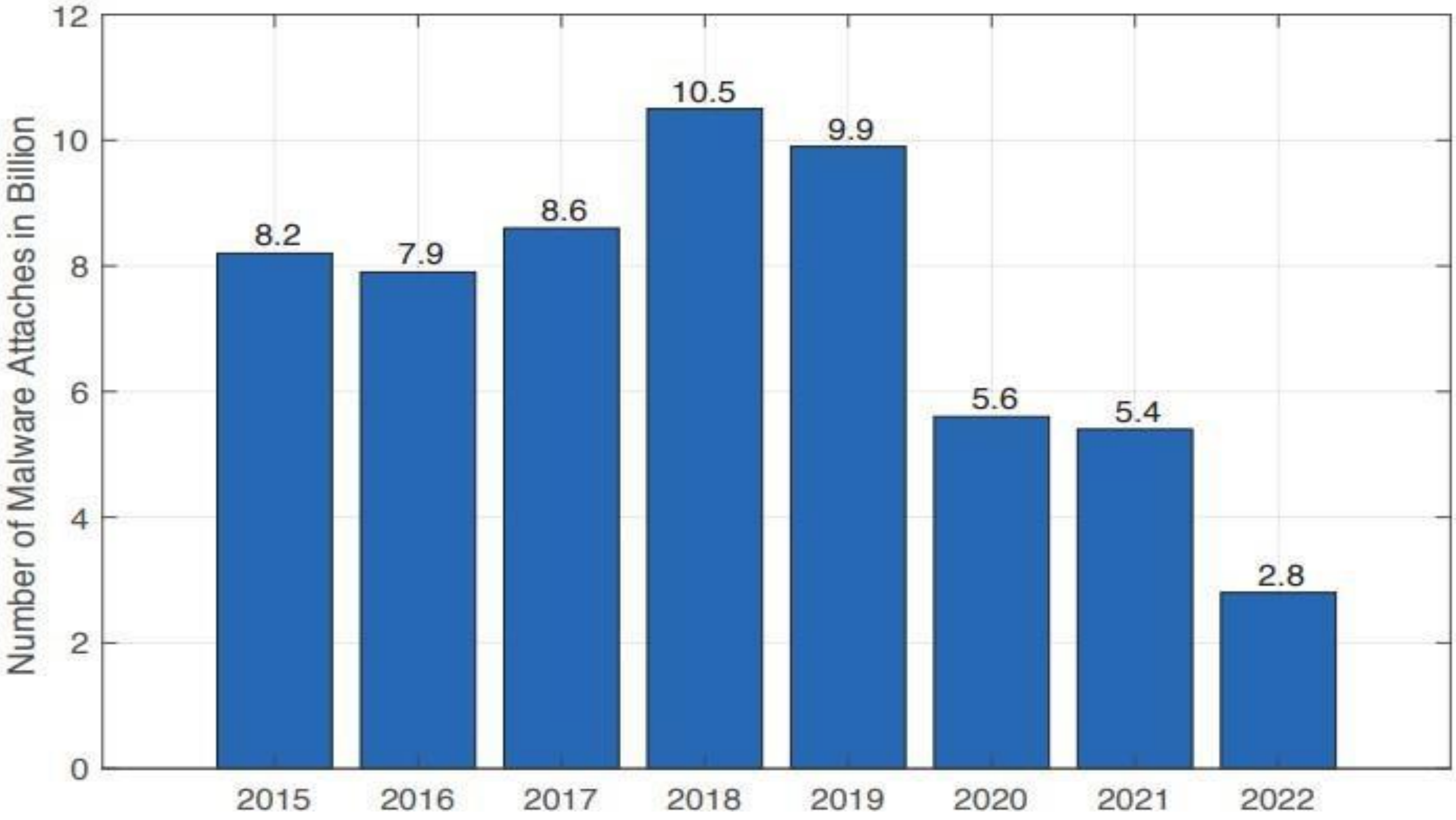



Fig: Malware Attacks From 2015 to 2022

# Problem Statement

- Generating negative illustrations have become a topic of great interest in the machine learning and cybersecurity research community.
- Researchers have worked extensively to make neural networks robust against adversarial attacks. However, many defenses fail as new or modified attack methods continue to bypass them.

# Objectives

- To implement a gradient based adversarial attack method on image based malware Classification systems.
- To evaluate the strength of vision transformers based malware classification model on powerful adversarial samples.

# Scope of Thesis

**Strength**

- Adversarial assaults on malware categorization methods will expose these models' flaws and make them more evasive and manipulable.

**Limitation**

- The research does not address how to mitigate the vulnerabilities revealed by FGSM and PGD assaults, since it lacks a defense mechanism.

# Potential Applications

- Testing Model Robustness
- Adversarial Training
- Studying Adversarial Behavior

# Literature Review[1]

| **Title** | **Date of Publication** | **Algorithm and Dataset** | **Results** | **Remarks** |
|---|---|---|---|---|
| Improving the Robustness of AI-Based Malware Detection Using Adversarial Machine Learning | 2021 | • Malevis Dataset<br>• Random Forest<br>• Efficient NetB0 | Efficient B0 was affected the least when compared to two models. | The use of adversarial training methods in the article increased the classifciation model’s robustness. |
| DefenseNet: A Resilent Network Against Adversarial Attacks On Recommendation Systems | 2021 | • Inception V3<br>• Resnet-50<br>• Vgg 16 | DefenseNet’s accuracy is higher than state of–the-art under adversarial attacks on Amazon benchmark dataset | DefenseNet ensemble architecture was designed in the study to create a recommender system that can defend against hostile attacks. |

# Literature Review[2]

| **Title** | **Date of Publication** | **Algorithm and Dataset** | **Results** | **Remarks** |
|---|---|---|---|---|
| GLEAM: GAN and LLM for Evasive Adversarial Malware | 2022 | • LLM (Large Language Model)<br>• GANs (Generative Adversarial Networks) | GLEAM achieved the goal by increasing the average evasion rate by 20.2% with GANs and 25.0% with LLMs, demonstrating the possibilities of expanded feature. | vulnerabilities in machine- learning-based malware detection algorithms(Logistic Regression, K-Nearest Neighbour, Random Forest, Decision Tree and Suport Vector Machine). |
| Prevention of DDoS Attacks Targeting Financial Services using Supervised Machine Learning and Stacked LSTM | 2022 | Stacked LSTM | Accuracy=99.3% | Stacked LSTM performed very well than othe neural networks like simple ANN,GRU in the detection of DDoS attacks based on the dataset collected from local banks of SriLanka. |

# Proposed Methodology[1]
## Theoritical  Background[1]

- **Traditional Malware Detection Techniques**
  - Signature-based Detection
  - Behavioral Analysis
  - Sandboxing

- **Intelligent Malware Detection**
  - Machine Learning-based
  - Deep Learning-based

# Proposed Methodology[2]
## Theoritical Background [2]

**Adversarial Attacks :** A "good" adversarial sample of an image is one that appears to be highly similar to the original but is misclassified by the network.

- White-box attacks
  - The attacker has full access to the model, including its architecture, parameters, and training data.

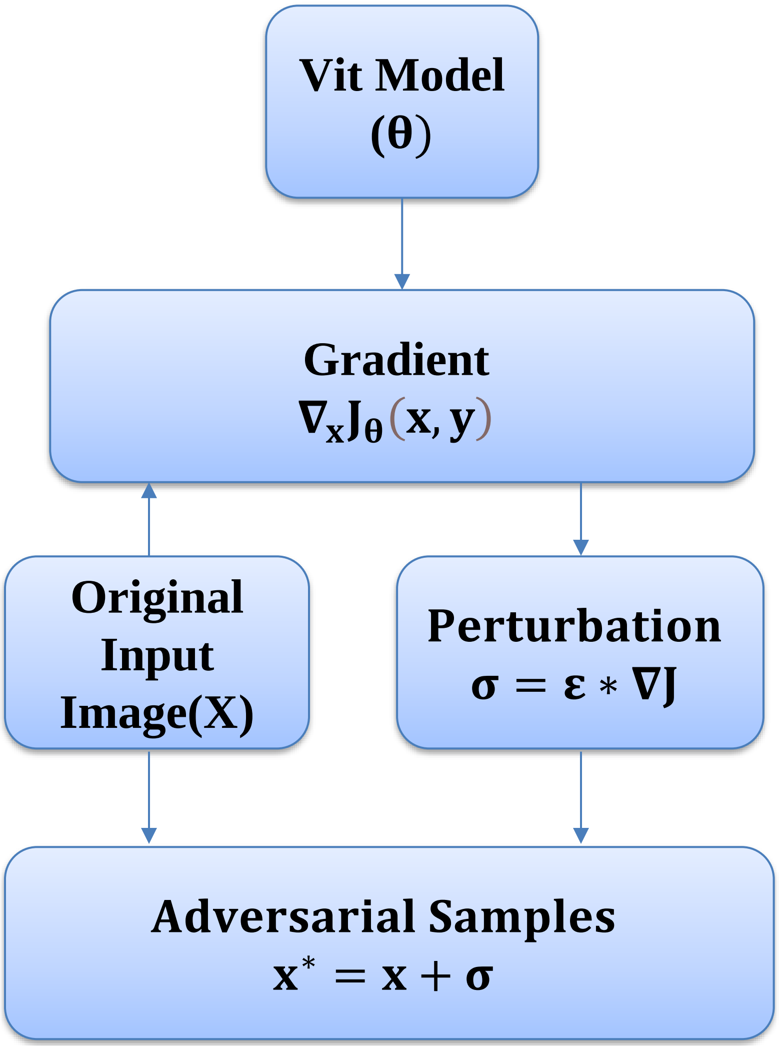
- Black-box attacks
  - The attacker has no access to the internal workings of the model but can observe the input-output behavior.

# Proposed Methodology [3]
## Theoritical  Background [3]

**Attack Generation Methods**

- FGSM
  - Simple, one-step
  - Can be used for quick evaluations of model robustness
- PGD
  - Complex, multi-step
  - More suitable for generating stronger adversarial examples

# Proposed Methodology[4]
## Mathematical Modelling[1]




- Perturbation is calculated using the gradients of the cost function J with respect to the input x.

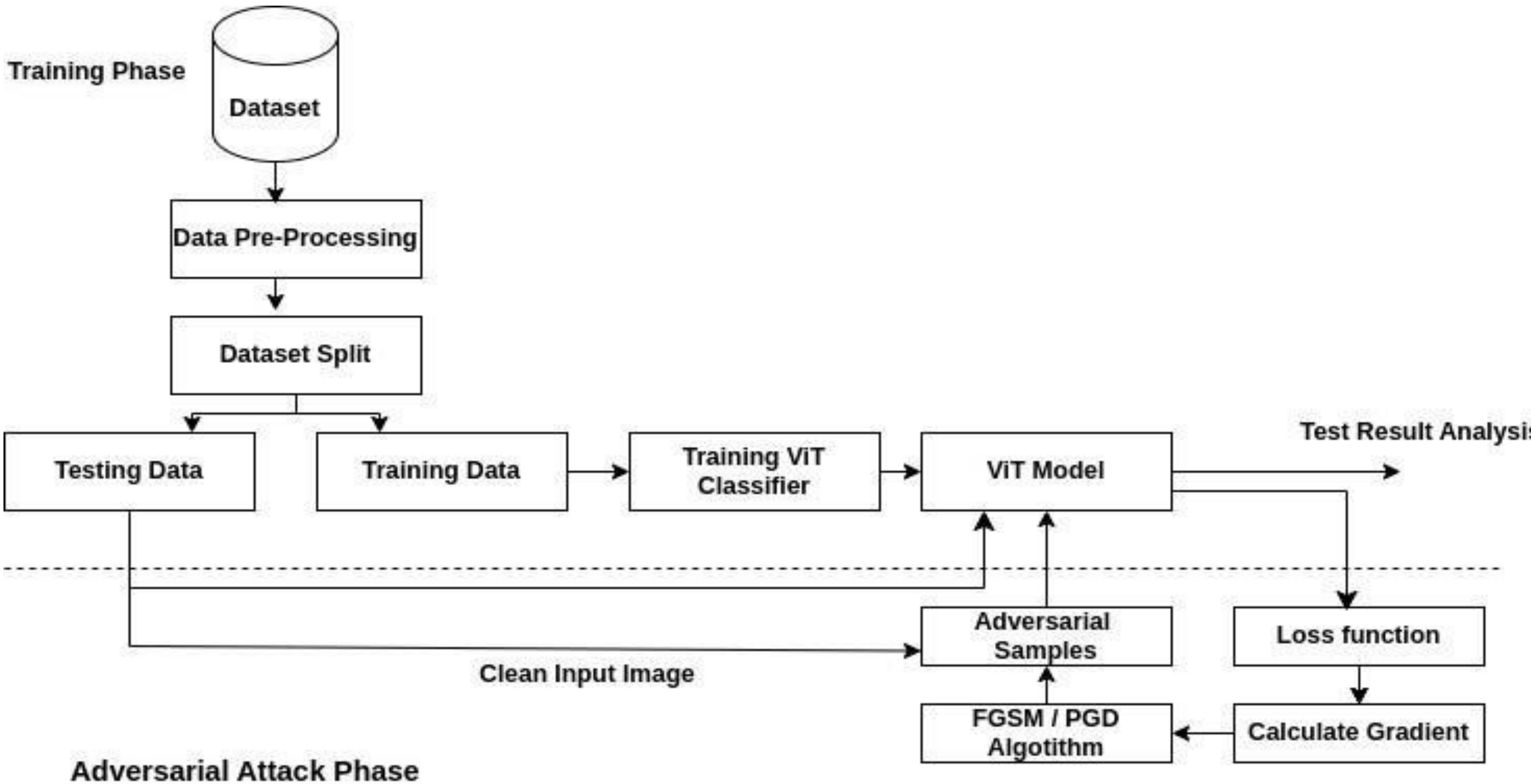
$$\sigma = \varepsilon * \nabla_x J_\theta(x, y)$$

- The adversarial example x is then calculated using :

$$x^* = x + \sigma$$

where
y is the label of x
θ is the parameter of the model
ϵ is tiny scalar value

# Proposed Methodology [5]
## System Flow Diagram



# Proposed Methodology[6]

## Dataset

Table : Malnet Dataset [16]

| S.N | Malware | Class(Original) | Family | Class(New) |
|---|---|---|---|---|
| 1 | Adware | 884k | 250 | 7k |
| 2 | Trojan | 179k | 441 | 7k |
| 3 | Benign | 79k | 1 | 7k |
| 4 | Riskware | 32k | 107 | 7k |
| 5 | Addisplay | 17k | 38 | 7k |

# Proposed Methodology [7]
## Working Principle [1]

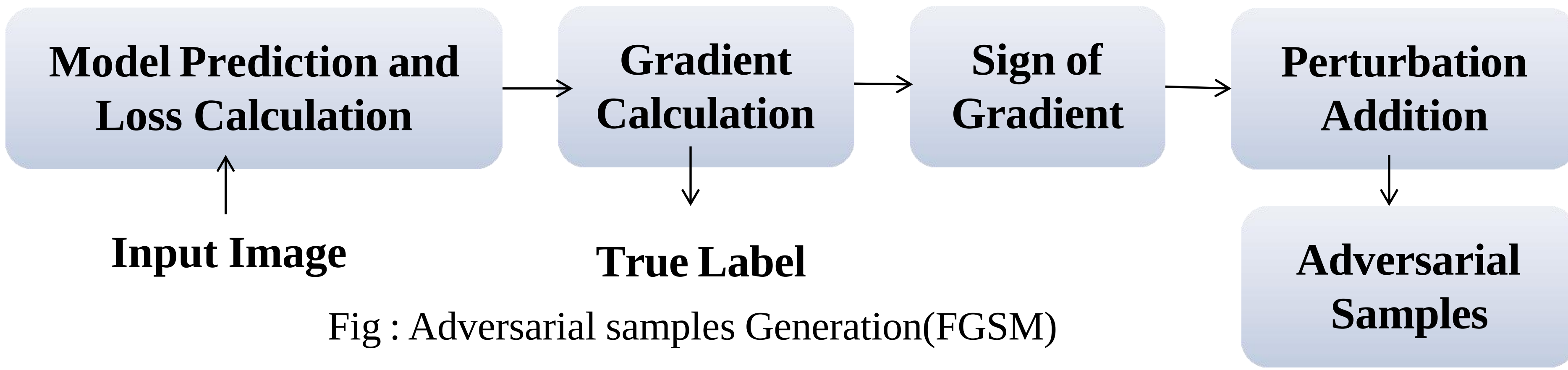



Fig : Adversarial samples Generation(FGSM)

- **Compute the Gradient**
  - Create the gradient of the loss function $J(\theta, x, y)$ with respect to the input x.

$$\nabla J(\theta, x, y)$$

- **Generate Perturbation**
  - Create a small perturbation using the sign of the computed gradient.

$$\delta = \in . sign(\nabla J(\theta, x, y))$$

- **Craft the Adversarial Samples**
  - Add the perturbation of the original input.

$$x' = x + \delta$$

# Proposed Methodology[8]

## Working Principle[2]

**Original Input Image(X)**

**Compute Gradient**
$\nabla_x J(\theta, x, y)$

**Update Adversarial Input**
$x_{t+1} = \Pi_e(x_t + \alpha \cdot \mathrm{sign}(\nabla_x J))$

**Project back to**
**ε-ball around x**

**Adversarial Example**
**x_adv**

where

- **$x$ :** Original input
- **$\nabla_x J(\theta, x, y)$ :** Gradient of the loss function with respect to input.
- **$\alpha$ :** Step size
- **$\varepsilon$ :** Maximum allowed perturbation .
- **$\Pi_e()$:** Projection operator to keep the adversarial image within $\varepsilon$ of the original image .
- **$x_t$ :** Iterative adversarial image at step *t*.

Fig :Adversarial samples Generation(PGD)

# Proposed Methodology [9]
## Working Principle [3]

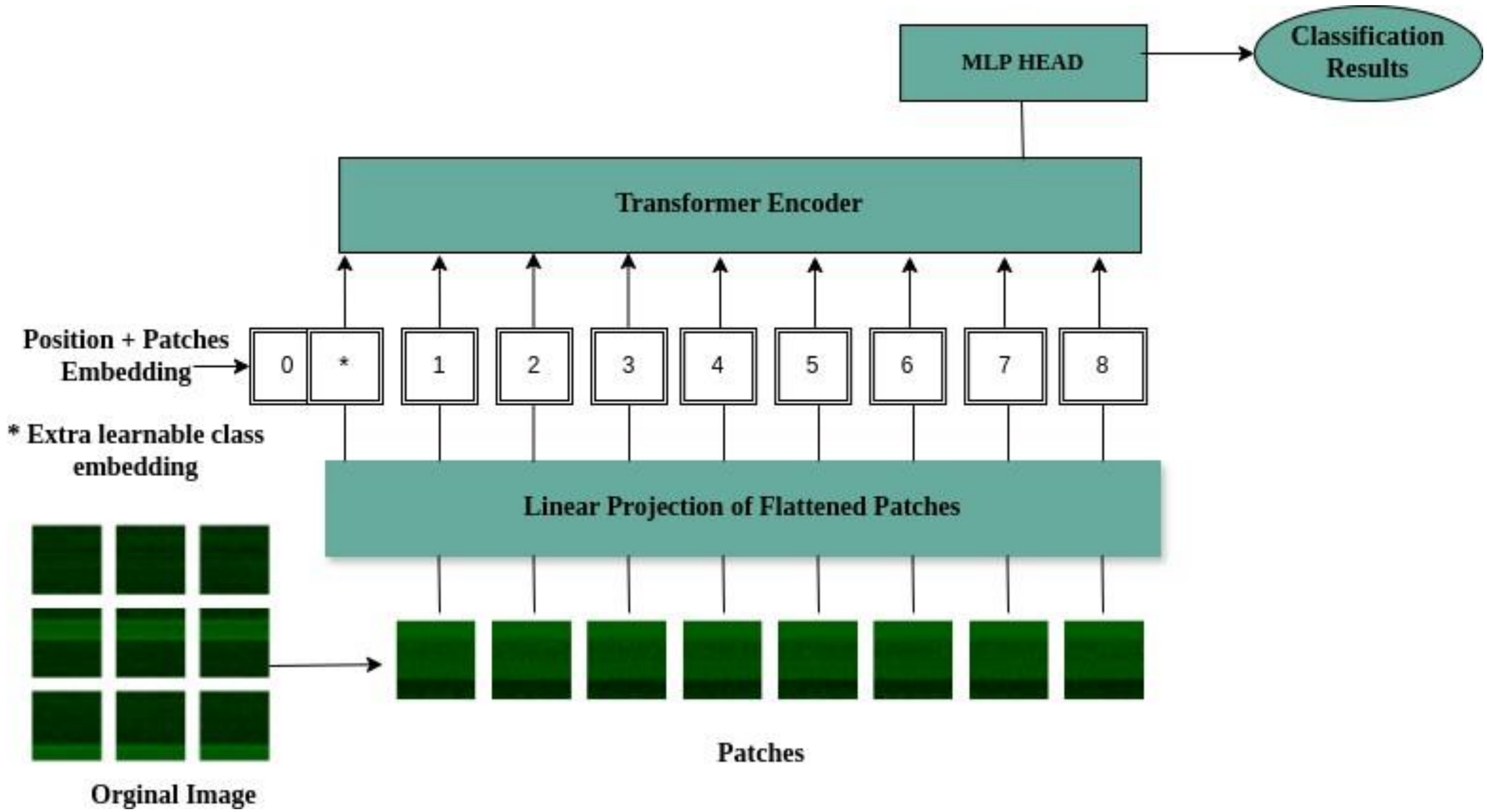



Fig : ViT Transformer Architecture

# Proposed Methodology[10]
## Working Principle [4]

- Patch Embedding
  - Split an input image into fixed-sized patches and flatten the patches.
- Learnable Positional Embedding
  - It  provide knowledge about the position of the input vectors.
  - It is necessary to retain 2D space information
  - Use learnable position embeddings instead of using fixed position embeddings (e.g. sin, cos).

# Proposed Methodology[11]
## Working Principle [5]

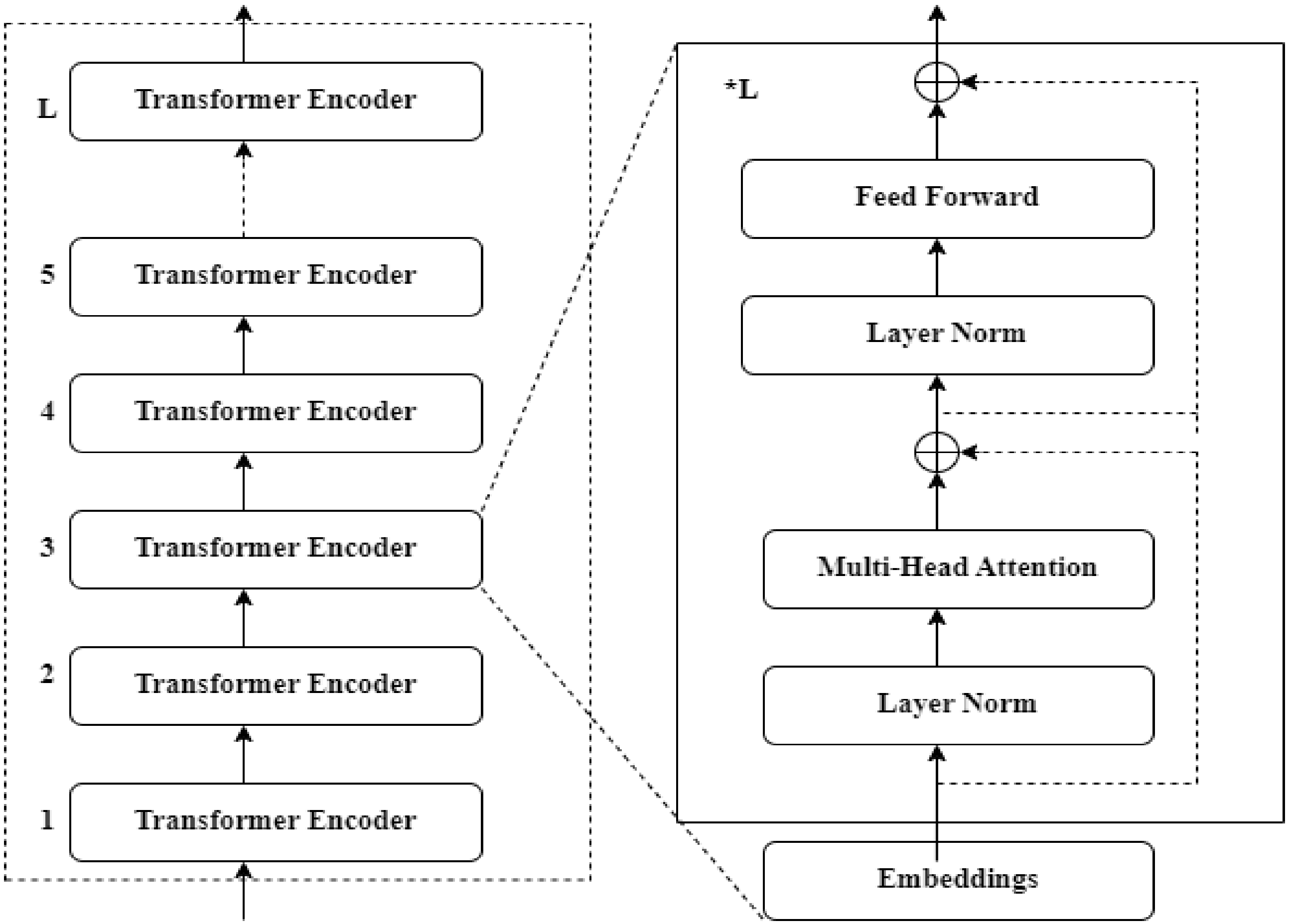



Fig: Transformer Encoder Block

# Proposed Methodology [12]
## Working Principle [6]

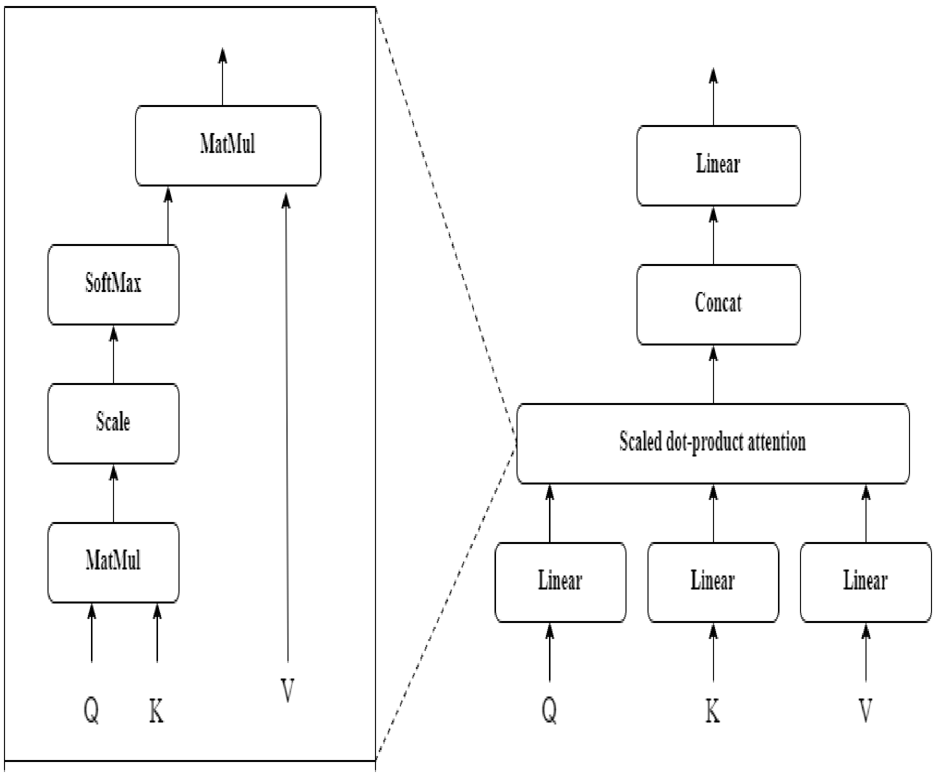



Fig: Multi-Head Self Attenuation

- **Query :** It represent the current element like image patch that is seeking information from other elements.
- **Key** : It represent the elements that contain information that might be useful to the query
- **Value :** The actual data associated with each key, which will be used to update the query.

# Proposed Methodology [13]
## Working Principle [7]

- **Input Projection :** Each head gets its own projected version of Q , K and V.

$$\boldsymbol{Q_i = QW_i^Q, \quad k_i = kW_i^k, \quad V_i = VW_i^v}$$

Where $W_i^Q, W_i^k, W_i^v$ are learnable weight matrices for head i

- **Scaled Dot-Product Attention (per Head)**

$$\textbf{Attention } (\boldsymbol{Q_i, k_i, V_i}) = \textbf{Softmax} (\frac{\boldsymbol{Q_i k_i^T}}{\sqrt{\boldsymbol{d_k}}})\boldsymbol{v_i}$$

□ Where $d_k$ : Dimension of Keys

- **Concatenation of Heads**

$$\textbf{Multihead(Q, K, V) = Concat(hea}\boldsymbol{d_1}\textbf{,hea}\boldsymbol{d_2}\textbf{,…,hea}\boldsymbol{d_h}\textbf{)}\boldsymbol{w^0}$$

Where $w^0$: Learnable output projection`matrix

- **Final Output :** The concatenated result is linearly transformed to produce the final multi-head attention output.

# Proposed Methodology[14]
## Working Principle[8]

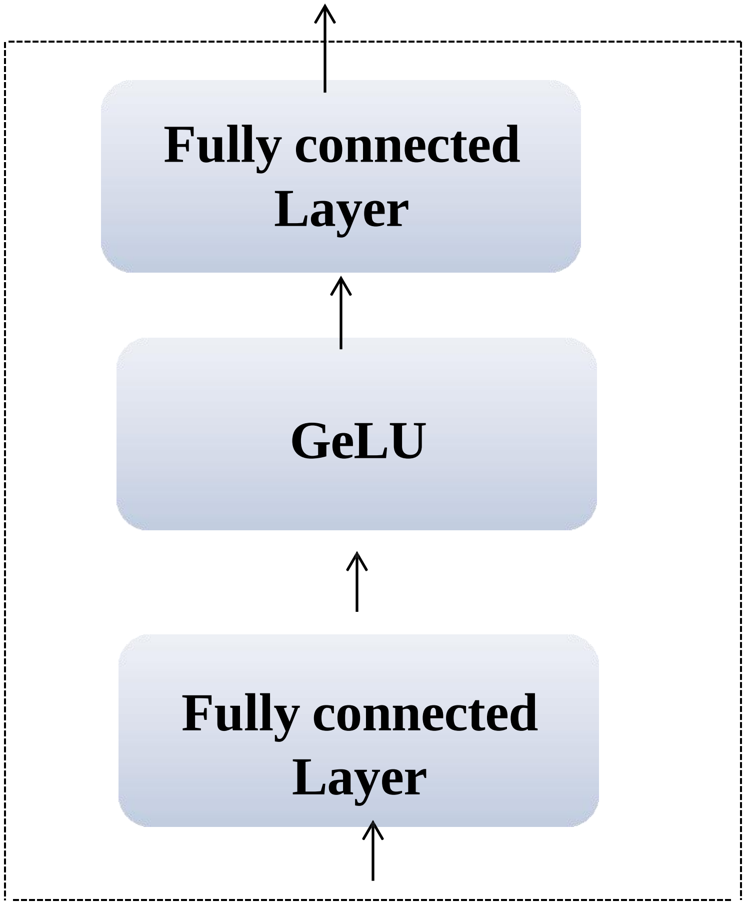



Fig : Multi-Layer Perceptron

# Results [1]
## Patch Embedding

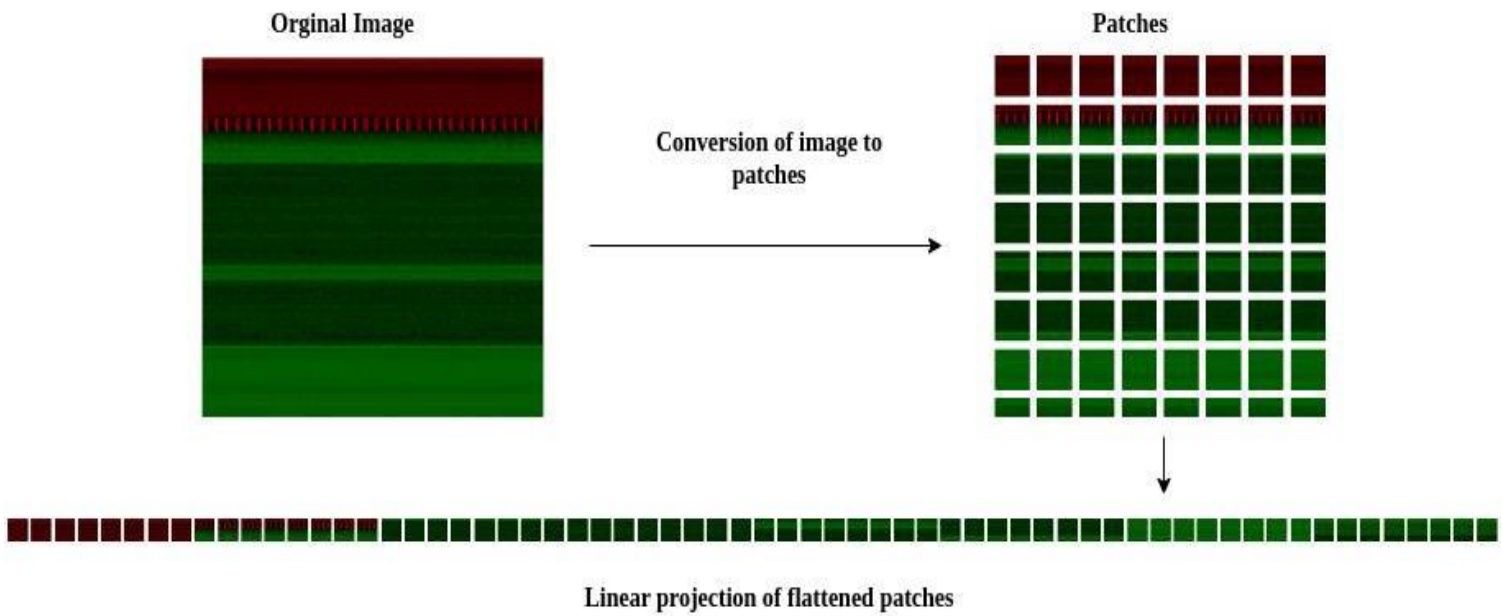



Fig : Process of Patch Embedding

# Results [2]
## Multi-Head Attention

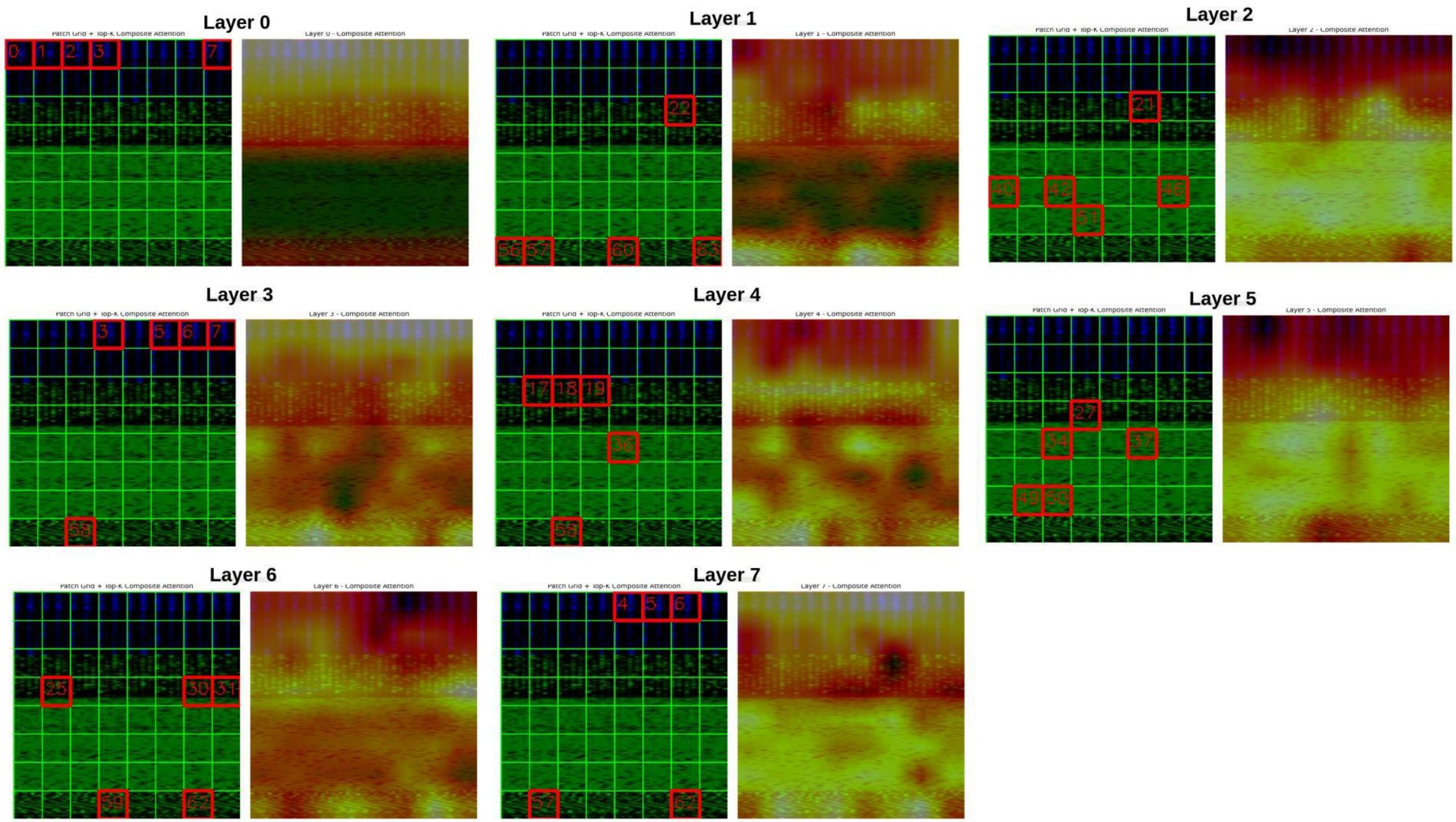
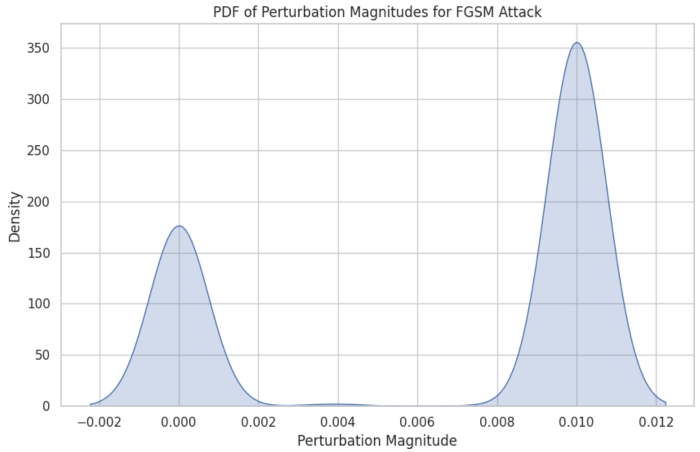



Fig : Multi-Head Attention Analysis Across Layers

# Results [3]
## PDF of Perturbation Magnitude



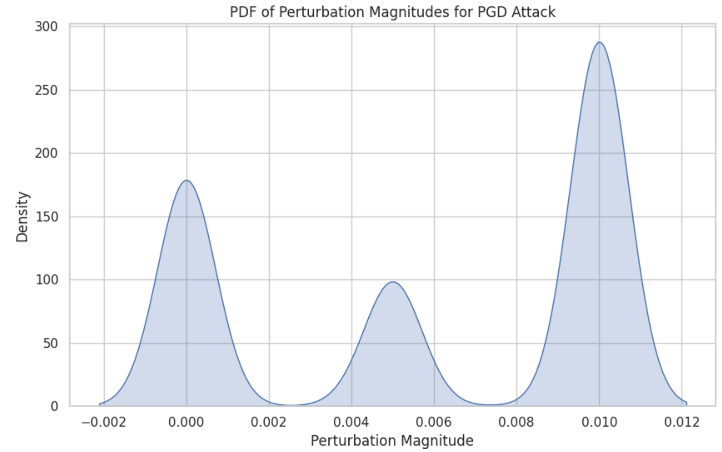




Fig : Comparision of Perturbation Magnitude for FGSM and PGD Attacks

| Feature | FGSM Perturbation | PGD Perturbation |
|---|---|---|
| Distribution Shape | Bimodal | Multimodal |
| Perturbation Spread | Uniform across pixels | Varies per pixel |
| Attack Complexity | Simple | Complex |

# Results [4]
## Generation of Adversarial Samples [1]

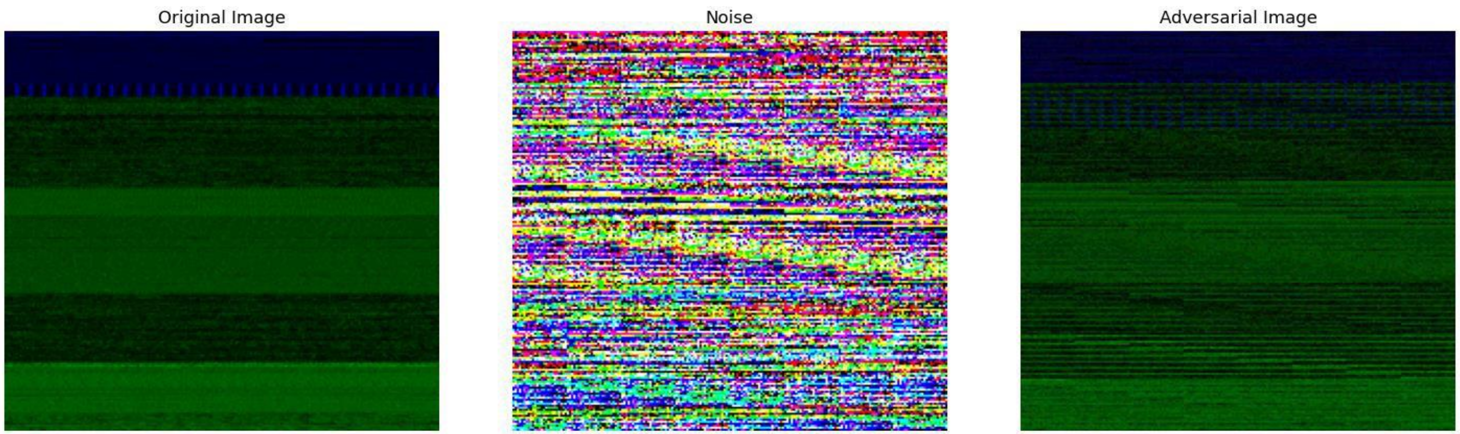



Fig : Generation of FGSM Samples at $\epsilon = 0.03$

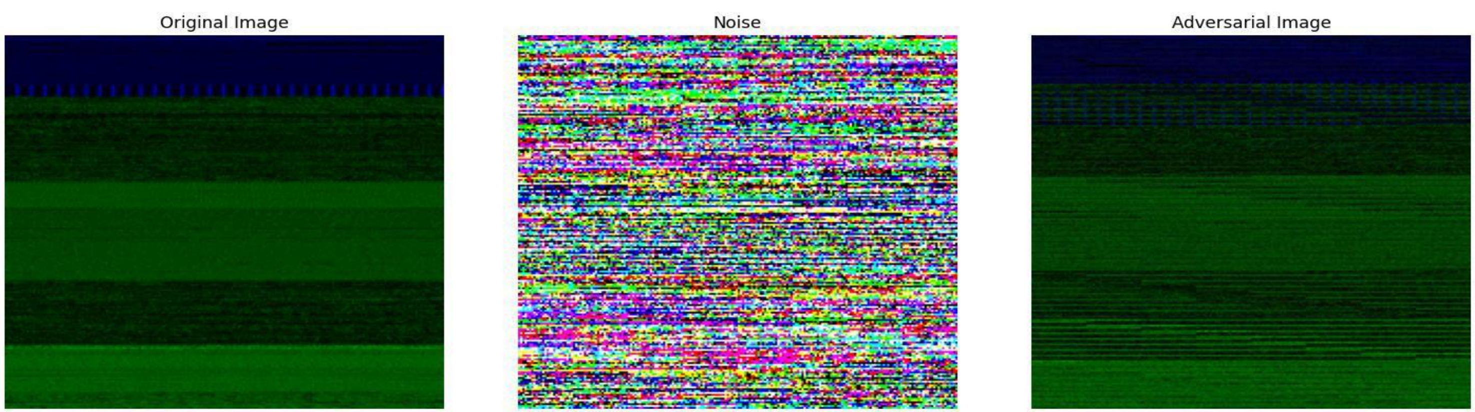



Fig : Generation of PGD Samples at $\epsilon = 0.03$

# Results [5]

## Generation of Adversarial Samples[2]

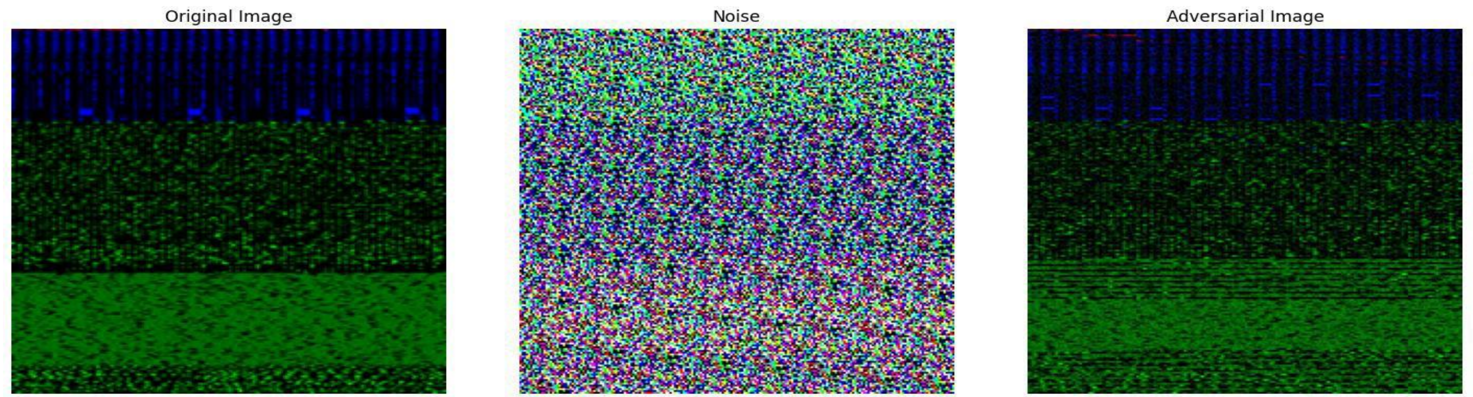



Fig : Generation of FGSM Samples at $\epsilon = 0.05$

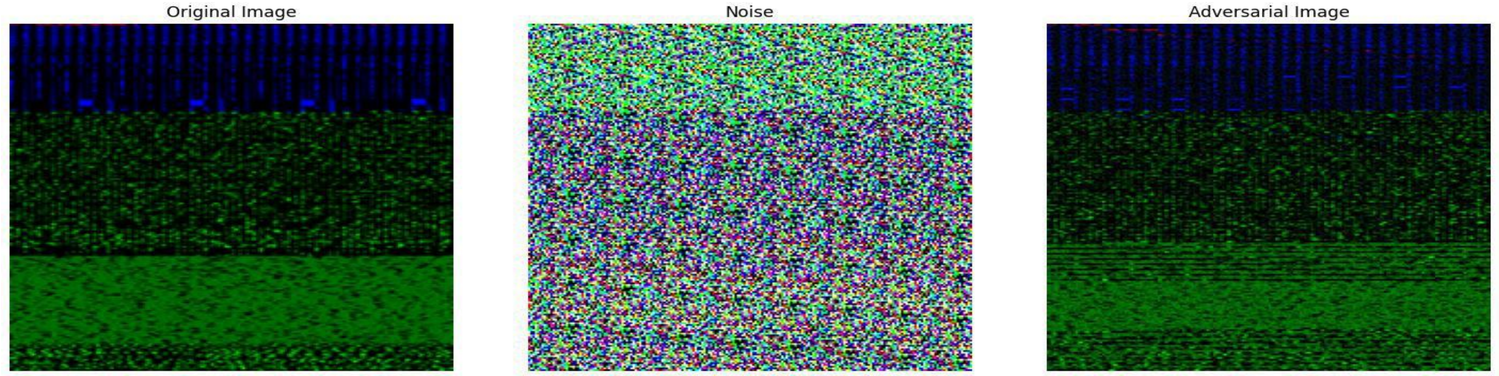



Fig : Generation of PGD Samples at $\epsilon = 0.05$

# Results [6]

## Confusion Matrix Before Attack [1]

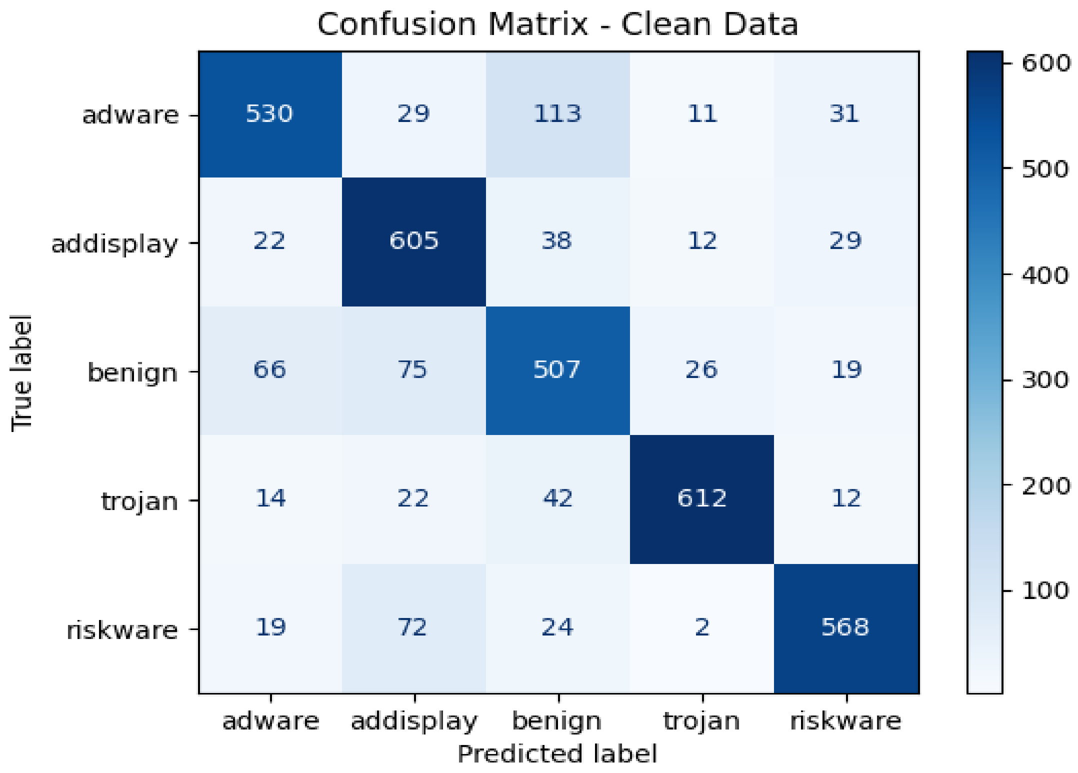



Fig : Confusion Matrix for Clean Image

# Results [7]

## Confusion Matrix After Attack [2]

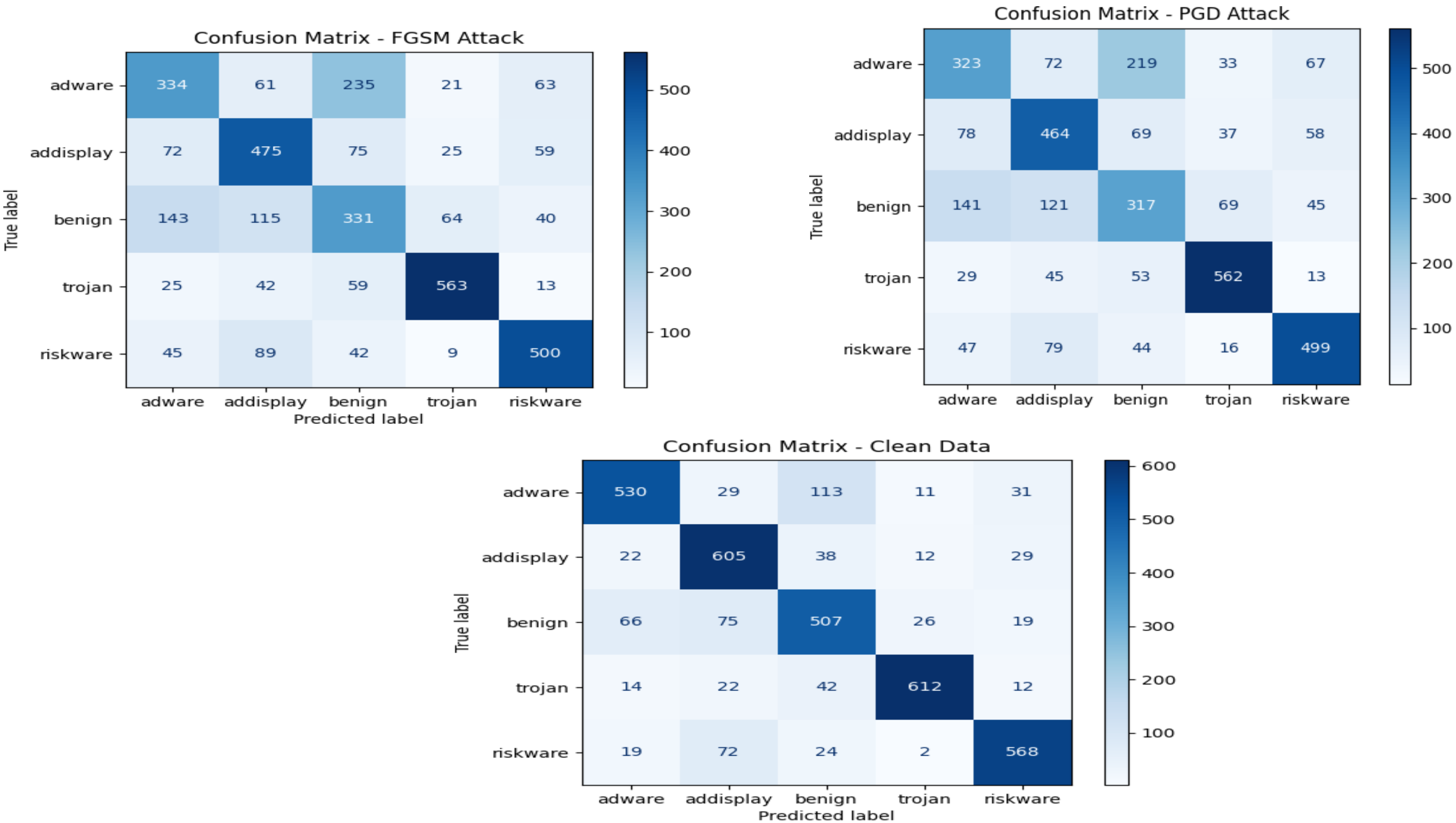



Fig : Comparision of Confusion Matrix Under FGSM and PGD Attacks at $\epsilon$= 0.001

# Results [8]

## Confusion Matrix After Attacks [3]

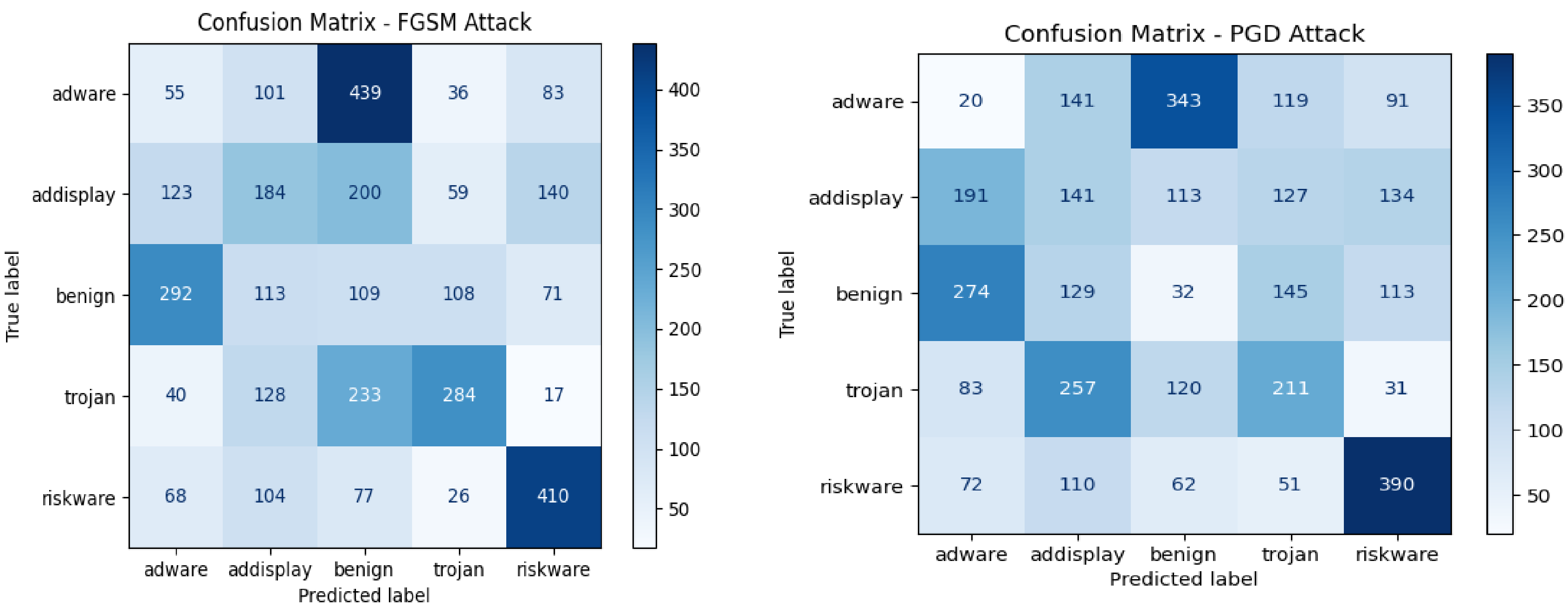



Fig : Comparision of Confusion Matrix Under FGSM and PGD Attacks at $\epsilon = 0.003$

# Results [9]

## Classification Report

Table 1: Classification Report of Clean Image

| Attacks | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Clean Image | 0.80 | 0.81 | 0.80 | 0.80 |

Table 2: Classification Report Under FGSM and PGD Attacks at $\epsilon = 0.001$

| Attacks | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| FGSM | 0.62 | 0.63 | 0.62 | 0.62 |
| PGD | 0.61 | 0.61 | 0.61 | 0.61 |

Table 3 : Classification Report Under FGSM and PGD Attacks at $\epsilon = 0.003$

| Attacks | Accuracy | Precision | Recall | F1-Score |
|---|---|---|---|---|
| FGSM | 0.30 | 0.31 | 0.30 | 0.30 |
| PGD | 0.22 | 0.21 | 0.22 | 0.22 |

# Results [10]

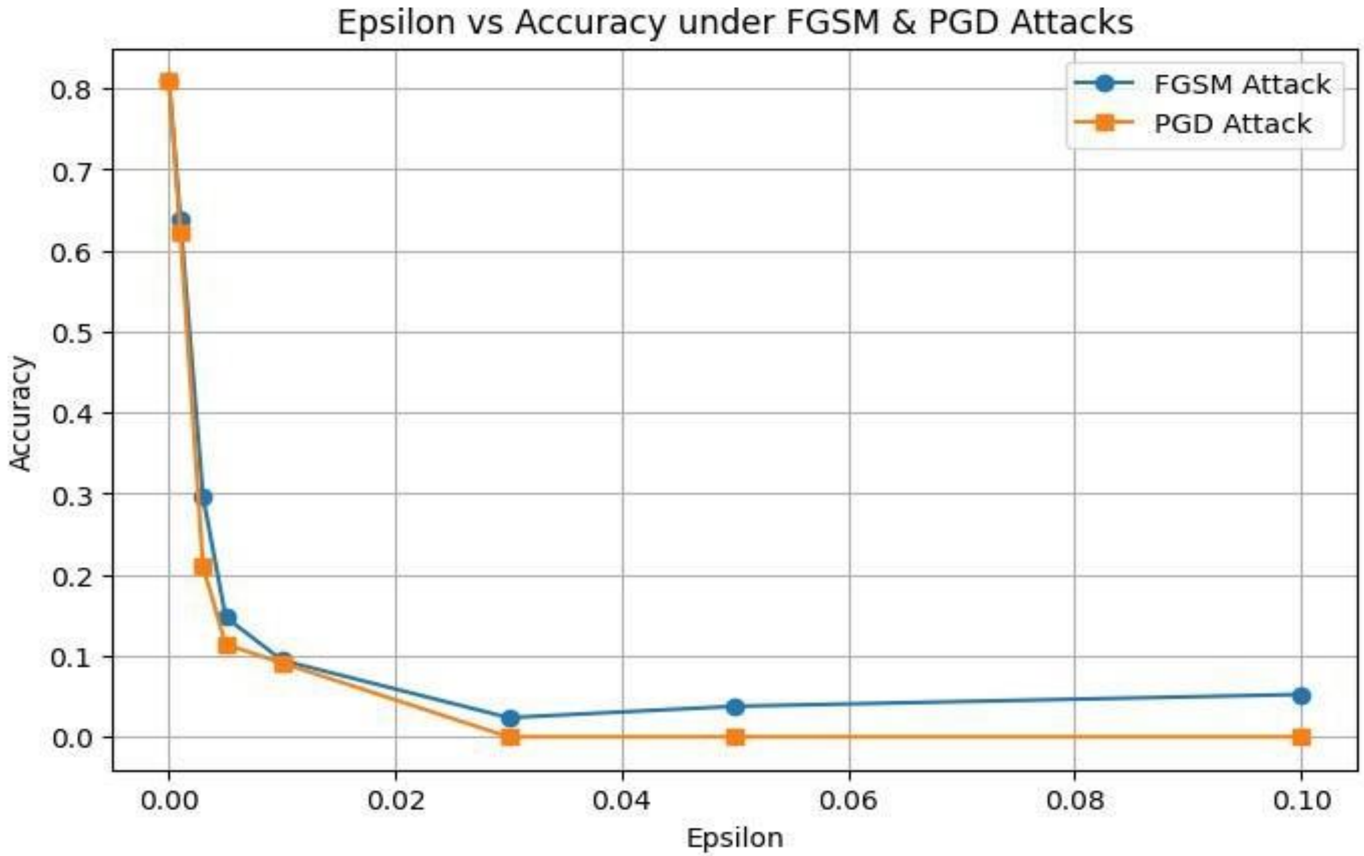



Fig : Accuracy vs Epsilon for FGSM and PGD Attacks at 40 Iteration

# Future Enhancement

- **Adversarial Defense Mechanisms :** To improve model resilience, use adversarial training by adding adversarial cases to the training procedure.
- **`Explainability and Visualization :** Analyze the effects of adversarial perturbations on various ViT attention heads using attention-based explainability methodologies.
- **Alternative Attack Strategies :** CW (Carlini-Wagner), DeepFool etc.
- **Hybrid Models for Robustness :** Investigate ensemble learning to successfully fight hostile perturbations by combining several models.
- **Real-World Deployment Considerations**

# Conclusion

- ViT-based malware classifiers suffer severe accuracy drops under adversarial attacks (21% with PGD, 30% with FGSM vs. 80% baseline).
- Despite advanced architecture and strong feature extraction, ViTs remain highly vulnerable, especially to iterative attacks like PGD.

# References[1]

[1] Ahmed Ali Mohammed Al-Saffar, Hai Tao, and Mohammed Ahmed Talab. Review of deep convolution neural network in image classification. In 2017 International conference on radar, antenna, microwave, electronics, and telecommunications (ICRAMET), pages 26–31. IEEE, 2017.

[2] Ahmed Bensaoud, Nawaf Abudawaood, and Jugal Kalita. Classifying malware im-ages with convolutional neural network models. International Journal of Network Security, 22(6):1022–1031, 2020.

[3] Gregory Conti, Erik Dean, Matthew Sinda, and Benjamin Sangster. Visual reverse engineering of binary and data files. In International Workshop on Visualization for Computer Security, pages 1–17. Springer, 2008.

[4] Scott Freitas, Rahul Duggal, and Duen Horng Chau. Malnet: A large-scale cyber-security image database of malicious software. arXiv preprint arXiv:2102.01072, 2021.

[5] Yuan Gao, Xiaojuan Sun, and Chao Liu. A general self-supervised framework for remote sensing image classification. Remote Sensing, 14(19):4824, 2022.

[6] Dan Hendrycks, Mantas Mazeika, Saurav Kadavath, and Dawn Song. Using self-supervised learning can improve model robustness and uncertainty. Advances in neural information processing systems, 32, 2019.

[7] Setia Juli Irzal Ismail, Hafiz Pradana Gemilang, Budi Rahardjo, et al. Self-supervised learning implementation for malware detection. In 2022 8th Inter-national Conference on Wireless and Telematics (ICWT), pages 1–6. IEEE, 2022.

# References[2]

[8] Manos Antonakakis, Tim April, Michael Bailey, Matt Bernhard, Elie Bursztein, Jaime Cochran, Zakir Durumeric, J Alex Halderman, Luca Invernizzi, Michalis Kallitsis, et al. Understanding the mirai botnet. In 26th {USENIX} security symposium ({USENIX} Security 17), pages 1093–1110, 2017.

[9] Setia Juli Irzal Ismail, Hafiz Pradana Gemilang, Budi Rahardjo, et al. Self-supervised learning implementation for malware detection. In 2022 8th Inter-national Conference on Wireless and Telematics (ICWT), pages 1–6. IEEE, 2022.

[10] Sachith Seneviratne, Ridwan Shariffdeen, Sanka Rasnayaka, and Nuran Kasthuriarachchi. Self-supervised vision transformers for malware detection. IEEE Access, 10:103121–103135, 2022.

[11] Yuan Gao, Xiaojuan Sun, and Chao Liu. A general self-supervised framework for remote sensing image classification. Remote Sensing, 14(19):4824, 2022.

[12] ] K. Jiang, P. Peng, Y. Lian, and W. Xu, “The encoding method of position embeddings in vision transformer”, Journal of Visual Communication and Image Representation, vol. 89, p. 103 664, 2022.

[13] Niket Bhodia, Pratikkumar Prajapati, Fabio Di Troia, and Mark Stamp. Transfer learning for image-based malware classification. arXiv preprint arXiv:1903.11551, 2019.

# References[3]


[14] N. Kathirkamanathan, B. Thevarasa, G. Mahadevan, N. Skandhakumar, and N. Kuruwitaarachchi, “Prevention of ddos attacks targeting financial services using supervised machine learning and stacked lstm”, in 2022 IEEE 7th International conference for Convergence in Technology (I2CT), IEEE, 2022, pp. 1–5.

[15] S. Jha, V. Luhach, and R. Poddar, “Retinal malady classification using ai: A novel vit-svm combination architecture”, in 2022 6th International Conference on Computing Methodologies and Communication (ICCMC), IEEE, 2022, pp. 1659–1664.

[16] Olorunjube James Falana, Adesina Simon Sodiya, Saidat Adebukola Onashoga, and Biodun Surajudeen Badmus. Mal-detect: An intelligent visualization approach for malware detection. Journal of King Saud University-Computer and Information Sciences, 34(5):1968–1983, 2022.

# Thank you